

Viruses, worms, and Trojan horses are all types of malicious software, commonly known as malware, but they differ in how they infect and affect computer systems:

- **Virus:**

- A computer virus is a self-replicating program or code that attaches itself to legitimate executable files or documents.
- When an infected file is executed, the virus activates and spreads to other files and, in some cases, to other computers or devices through infected files.
- Viruses often have a payload, which can be destructive or harmful, such as deleting files, corrupting data, or stealing information.
- Examples of well-known computer viruses include the Melissa virus and the ILOVEYOU virus.

- **Worm:**

- A computer worm is a self-contained, self-replicating program that does not need to attach itself to other files to spread.
- Worms exploit vulnerabilities in networked systems to replicate and spread automatically to other computers or devices over a network or the internet.
- They can cause network congestion, degrade system performance, or carry out malicious actions, such as installing backdoors for remote control or stealing sensitive data.
- Notable examples of computer worms include the Conficker worm and the WannaCry ransomware worm.

- **Trojan Horse (Trojan):**

- A Trojan horse is a type of malware disguised as a legitimate or benign program or file to deceive users into running or installing it.
- Unlike viruses and worms, Trojans do not self-replicate but rely on social engineering to trick users into executing them.
- Once activated, Trojans can perform various malicious actions, including spying, data theft, remote control, or delivering additional malware.
- Common examples include Trojans that pretend to be useful software or games but actually carry out harmful activities.